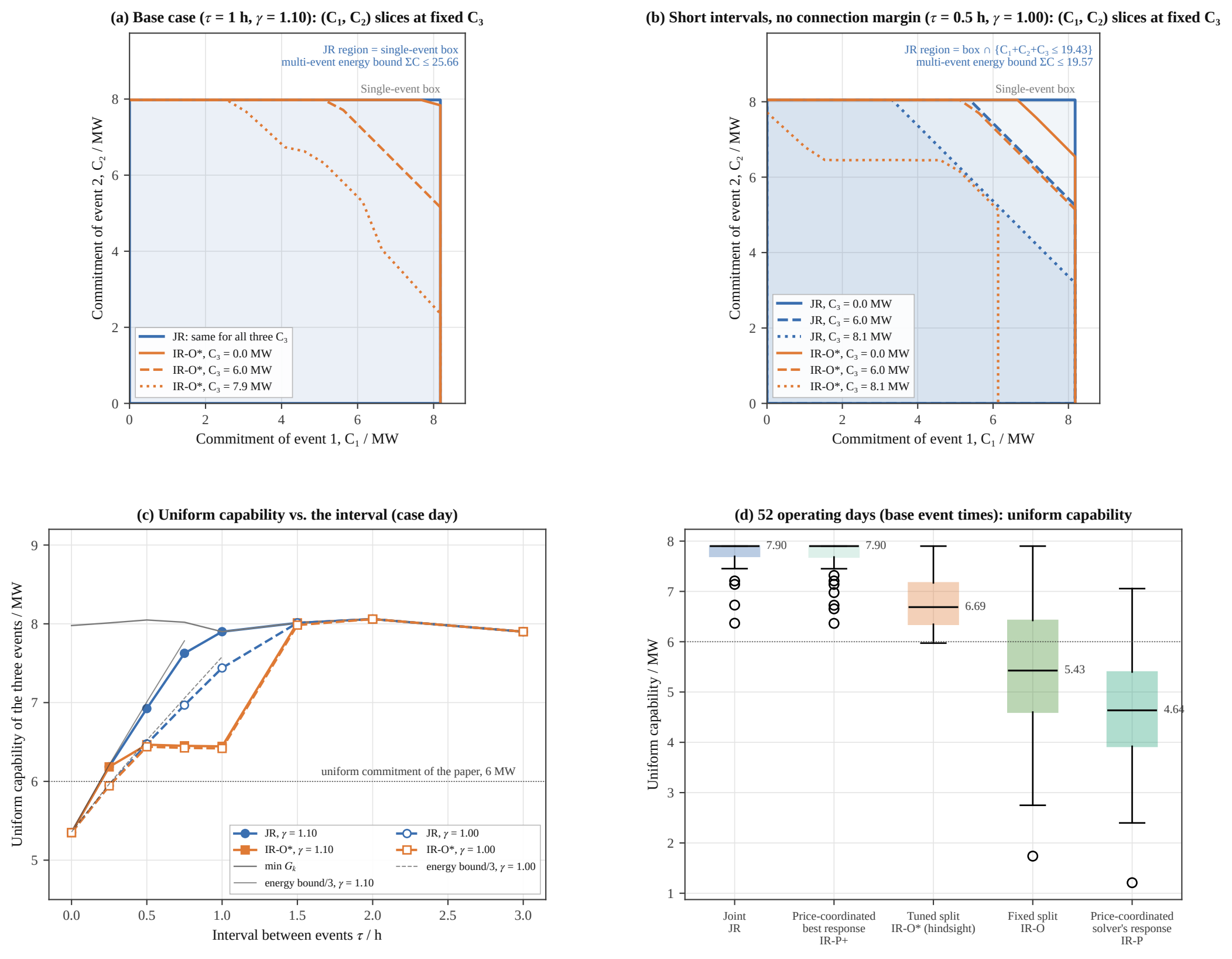
(a) Base case (τ = 1 h, γ = 1.10): (C_1, C_2) slices at fixed C_3
JR region = single-event box
multi-event energy bound ΣC ≤ 25.66
Single-event box
JR: same for all three C_3
IR-O*, C_3 = 0.0 MW
IR-O*, C_3 = 6.0 MW
IR-O*, C_3 = 7.9 MW
Commitment of event 2, C_2 / MW
Commitment of event 1, C_1 / MW
(b) Short intervals, no connection margin (τ = 0.5 h, γ = 1.00): (C_1, C_2) slices at fixed C_3
JR region = box ∩ {C_1+C_2+C_3 ≤ 19.43}
multi-event energy bound ΣC ≤ 19.57
Single-event box
JR, C_3 = 0.0 MW
JR, C_3 = 6.0 MW
JR, C_3 = 8.1 MW
IR-O*, C_3 = 0.0 MW
IR-O*, C_3 = 6.0 MW
IR-O*, C_3 = 8.1 MW
Commitment of event 2, C_2 / MW
Commitment of event 1, C_1 / MW
(c) Uniform capability vs. the interval (case day)
uniform commitment of the paper, 6 MW
JR, γ = 1.10
IR-O*, γ = 1.10
min G_k
energy bound/3, γ = 1.10
JR, γ = 1.00
IR-O*, γ = 1.00
energy bound/3, γ = 1.00
Uniform capability of the three events / MW
Interval between events τ / h
(d) 52 operating days (base event times): uniform capability
7.90
7.90
6.69
5.43
4.64
Uniform capability / MW
Joint JR
Price-coordinated best response IR-P+
Tuned split IR-O* (hindsight)
Fixed split IR-O
Price-coordinated solver's response IR-P